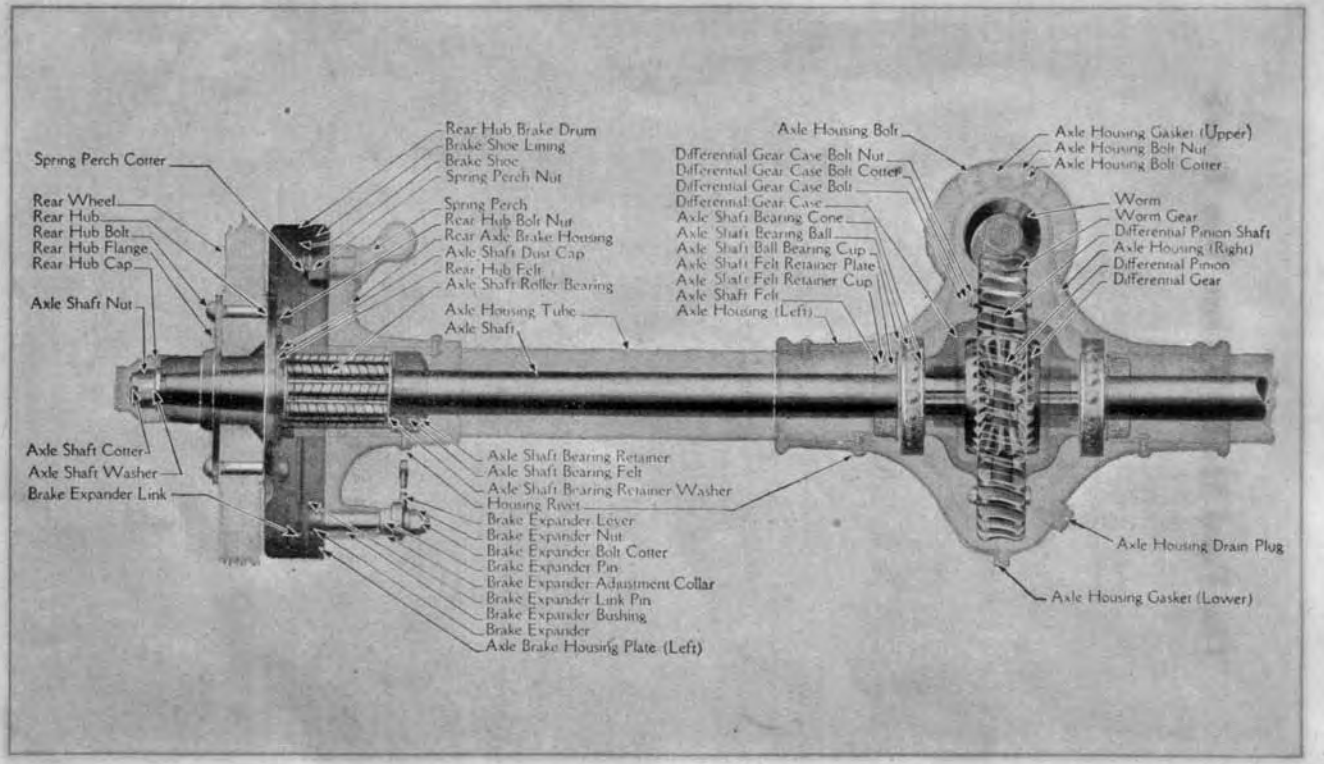
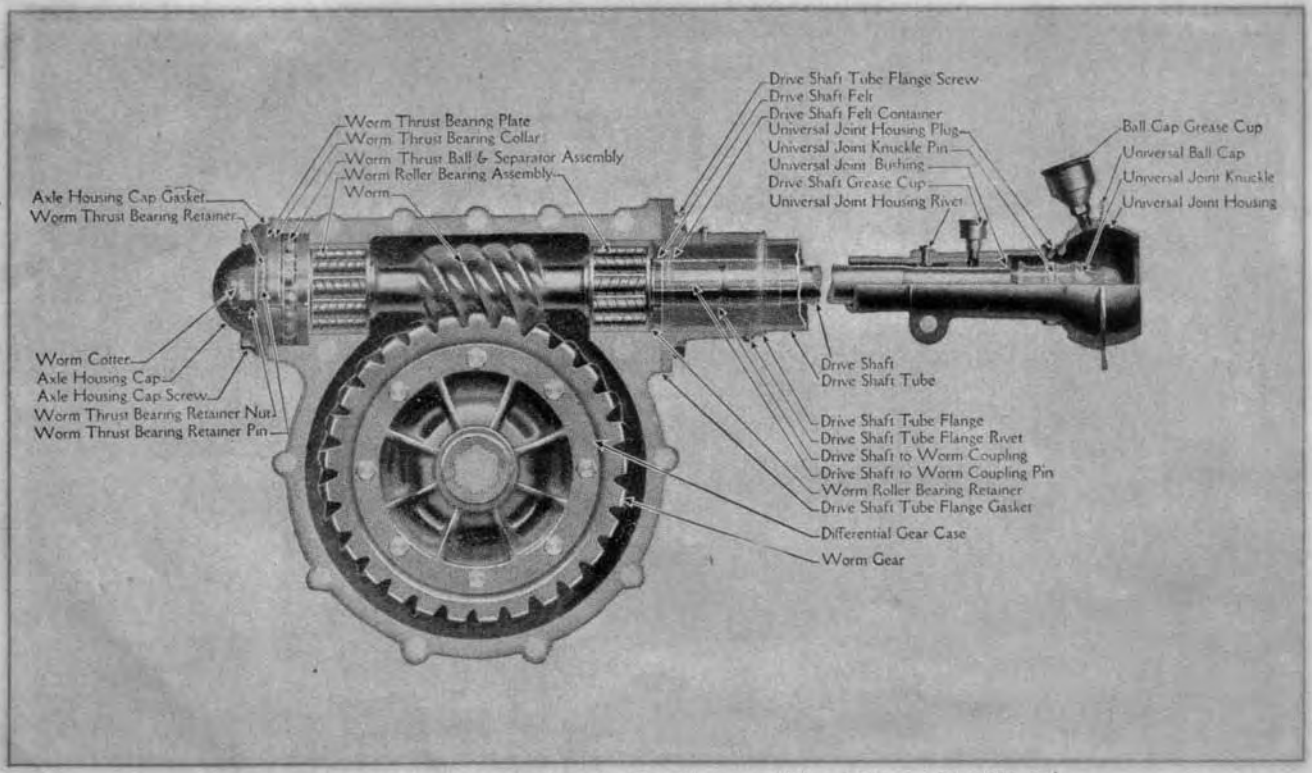
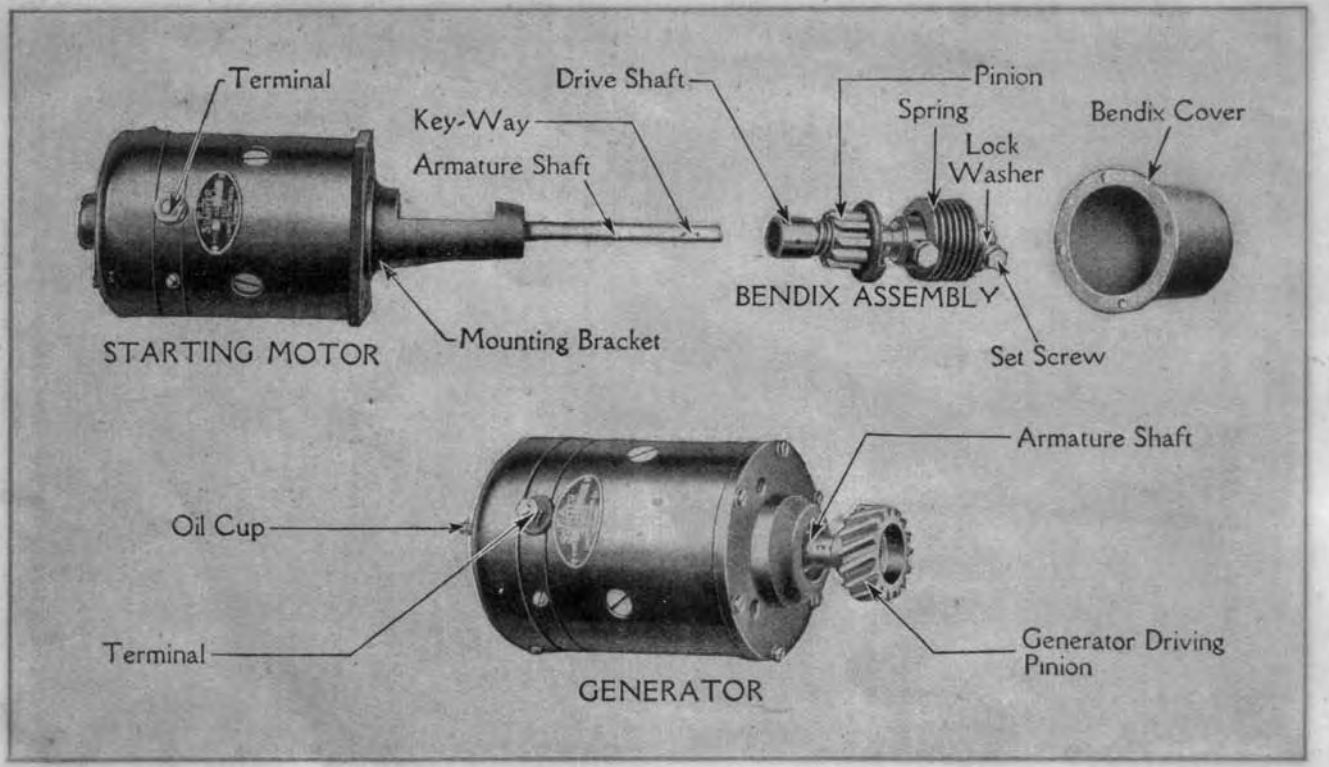


Truck Rear Axle—Longitudinal View. (Cut No. 20)



Truck Rear Axle—Cross Section showing Worm and Worm Gear. (Cut No. 21)



Starter and Generator Units. (Cut No. 22)

The Ford Starting and Lighting System

Of what does the Starting and Lighting System consist? — *Answer No. 117*

The starting and lighting system is of the two unit type and consists of the starting motor, generator, storage battery, a nmeter, and lights, together with the necessary wiring and connections.

Where is the Starter located? — *Answer No. 118*

The starting motor is mounted on the left-hand side of the engine and bolted to the transmission cover. When in operation the pinion on the Bendix drive shaft engages with the teeth on the flywheel.

What if the Engine fails to start? — *Answer No. 119*

If the starting motor is turning the crank shaft over and the engine fails to start, the trouble is not in the starting system. In this event release the button at once so as not to unnecessarily discharge the battery and inspect the carburetor and ignition system to determine the trouble.

What if the Starting Motor fails to act? — *Answer No. 120*

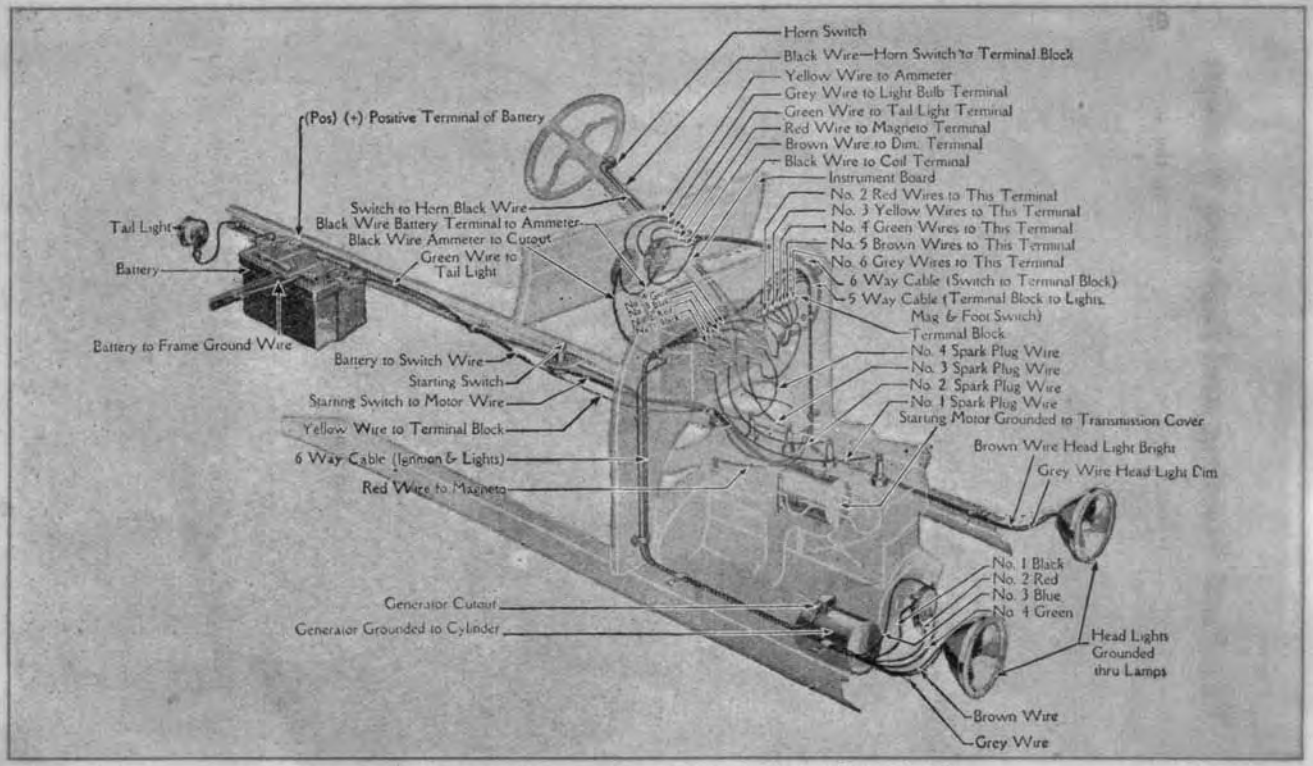
If the starting motor fails to act, after pushing the button, first inspect the terminal on the starting motor, the two terminals on the battery and the two terminals on starting switch, making sure all of the connections are tight; then examine the wiring for a break in the insulation that would cause a short circuit. If the wiring and connections are O. K. and the starting motor fails to act, test the battery with a hydrometer. If the hydrometer reading is less than 1.225 the trouble is no doubt due to a weak or discharged battery.

What if the driver steps on Starting Button when engine is running? — *Answer No. 121*

Should the driver accidentally step on the starting button while the engine is running, no harm will result. The pinion merely touches the revolving flywheel gear once and immediately rotates with the threaded shaft out of contact with the flywheel, in the same manner as when it has been disengaged by the engine's starting.

How is the Generator operated? — *Answer No. 122*

The generator is mounted on the right-hand side of the engine and bolted to the cylinder front end cover. It is operated by the pinion on the armature shaft engaging with the large time gear. The charging rate of generator is set so as to cut in at engine speeds corresponding to 10 miles per hour in high speed and reaches a maximum charging rate at 20 miles per hour. At higher speeds the charge will taper off, which is a settled characteristic of a generator. This operation of cutting in and cutting out at suitable speeds is accomplished by the cut-out, which is mounted on the generator. This cut-out is set properly at the factory and should not under any circumstances be tampered with.



Wiring Diagram Cars Equipped with Starter. (Cut No. 23)

What about Oiling? **Answer No. 123**

The starting motor is lubricated by the Ford splash system, the same as the engine and transmission. The generator is lubricated by a splash of oil from the time gears. In addition an oil cup is located at the end of the generator housing and a few drops of oil should be applied occasionally.

What should be done when repairing the Ignition? **Answer No. 124**

The introduction of a battery current into the magneto will discharge the magnets and whenever working on the ignition system or wiring do not fail to disconnect the positive wire from the battery. The end of this wire should be wound with tape to prevent its coming in contact with the terminal again.

How does the Ammeter operate? **Answer No. 125**

The ammeter is located on the instrument board. This indicator registers "charge" when the generator is charging the battery and "discharge" when the lights are burning and the engine not running above 10 miles per hour. At an engine speed of 15 miles per hour or more the ammeter should show a reading of from 10 to 12. If the engine is running above 15 miles per hour and the ammeter does not show a proper reading, first inspect the terminal posts on the ammeter, making sure that the connections are tight, then disconnect the wire from the terminal on generator, and with the engine running at a moderate speed, take a pair of pliers or a screw-driver and short-circuit the terminal stud on the generator to the generator housing. If the generator is O. K., a good live spark will be noted. (Do not run the engine any longer than is necessary with the terminal wire disconnected.) Next inspect the wiring from the generator through the ammeter, to the battery for a break in the insulation that would result in a short-circuit. If the trouble is not located, then remove the dust cap from the end of generator and thoroughly clean the generator commutator, using for this work a fine grade of sandpaper which has been slightly oiled. With the motor running hold the sandpaper against the commutator with the fingers until all dirt has been removed and a bright surface attained.

How are the Lights operated? **Answer No. 126**

The lighting system consists of two headlights and a tail light operated by a combination lighting and ignition switch located on the instrument board. The headlamp bulbs are of 6-8 volt, double filament type. The major filament is 18 candle-power, and the minor filament is 2¾ candle-power. The small bulb used in the tail light is of 6-8 volt, single contact, two candle-power type. All of the lamps are connected in parallel so that the burning out or removal of any one of them will not affect the other. Current for the lamps is supplied by the battery. Do not connect the lights to the magneto as it will result in burning out the bulbs and might discharge the magnets. Cut No. 23 shows the different circuits and the course of the current.

What about repairing Starter and Generator **Answer No. 127**

If either the starter or generator fails to give proper service, the owner should at once consult an authorized Ford dealer. ***Owners should not attempt to repair or adjust the mechanism of the starter and generator.***

How is the Starter removed?

Answer No. 128

When removing the starter to replace transmission bands, or for any other reason, first remove the engine pan on the left-hand side of the engine and with a screw-driver remove the four small screws holding the shaft cover to the transmission cover. Upon removing cover and gasket, turn the Bendix drive shaft around so that the set screw on the end of the shaft is at the top. Immediately under the set screw is placed a lock washer, designed with lips or extensions opposite each other. One of these is turned against the collar and the other is turned up against the side of the screw head. Bend back the lip which has been forced against the screw and remove the set-screw. As the lock washer will no doubt be broken or weakened in removing the starter, a new one must be used when replacing it. Next, pull the Bendix assembly out of the housing, being careful that the small key is not misplaced nor lost. Remove the four screws which hold the starter housing to the transmission cover, and pull out the starter, taking same down through the chassis—this is why it was necessary to remove the engine pan. Extreme care should be used in removing the Bendix drive and other parts that none are misplaced nor lost and that they are replaced in their former position. In replacing the starter, be sure that the terminal connection is placed at the top. If the car is to be operated with the starter removed, be sure to put the transmission cover plates in position. These plates may be obtained from the nearest dealer.

How is the Bendix Drive assembled to the Starting Motor?

Answer No. 129

When assembling the Bendix Drive to the starting motor shaft, care must be used to see that the stop nut or bearing which enters the mounting bracket on the starting motor is not too tight; also that the bearing is in proper alignment with the bracket. The bearing should be oiled and then fitted so that it can be turned readily with the fingers. If the bearing is too tight, it should be dressed down with an oil stone. Too tight a fit will cause the bearing to freeze to the bracket, resulting in serious damage to the starter.

How is the Generator removed?

Answer No. 130

If it is found necessary to remove the generator, first take out the three cap screws holding it to the front end cover and by placing the point of a screw-driver between the generator and front end cover, the generator may be forced off the engine assembly. Always start at the top of the generator and force it backward and downward at the same time. Plates may be obtained from the nearest dealer to place over the time gear if the car is to be operated with the generator removed.

What should be done when replacing the Generator?

Answer No. 131

When installing the generator, the drive pinion must be properly meshed with the large time gear. The generator bracket, that is, the section to which the generator is bolted is separate from the cylinder block and the meshing of the generator driving pinion with the large time gear can be regulated by the use of one or more paper gaskets between the bracket and the cylinder block. The bracket should rest tightly on the crankcase gasket and line up with the face of the time gear case. If these gears are meshed too tightly, a humming noise will result, also the generator shaft will be thrown out of alignment.

Can the Engine be run with the Generator disconnected from the battery?

Answer No. 132

If for any reason the engine is run with the generator disconnected from the battery, as on a block test, or when battery has been removed for repair or recharging, be sure that the generator is grounded by running a wire from the terminal on generator to one of the dust cover screws in the yoke. Two strands of shipping tag wire may be used for this purpose. Be sure that the connections at both ends of the wire are tight. Failure to do this when running the engine with the generator disconnected from the battery will result in serious injury to the generator. ***Never ground the generator through the cut-out.***

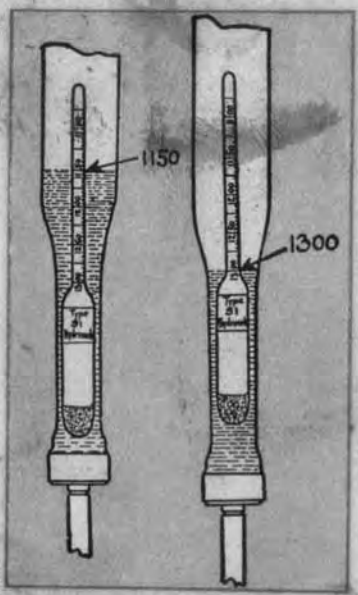
What type of Battery is used?

Answer No. 133

The Ford Starting System uses a six-volt, three-cell battery.

How are Hydrometer Readings taken?

Answer No. 134



Hydrometer Readings. (Cut No. 24)

Hydrometer readings should be taken about every two weeks to make sure that the generator is keeping the battery charged. To take a hydrometer reading, remove the filling plugs (remove the plug from only one cell at a time). Next, insert the hydrometer syringe in the filler tube and draw up enough of the solution to float the glass bulb inside the instrument. The reading of the scale at the surface of the liquid (see Cut No. 24) gives the strength of the solution. Be sure to return the electrolyte to the cell from which it was taken. Following is a list of the readings with their indications. The readings in parentheses apply to batteries used in tropical climates where water never freezes. Readings of 1.275 (1.200) or more indicate a fully charged battery. Readings of less than 1.225 (1.130) but more than 1.150 (1.080) indicate complete discharge. Hydrometer tests taken immediately after filling with water and before it has become thoroughly mixed with the electrolyte will not show the true condition of the battery. If the hydrometer reading shows the battery less than one-half charged, it should be taken to the nearest authorized Battery Service Station for recharging. Continued operation in a less than half-charged condition is injurious to the battery somewhat as running on a soft or deflated tire is injurious to the tire. Before replacing the battery, the cause of the discharged condition should be removed. It may be due to leaks or grounds in the car wiring or to the electric system having gotten out of adjustment so that the battery is not kept supplied with the proper amount of current from the generator. If the reading in one cell is more than

50 points different from the others, it indicates that the cell is not in good order and the battery should be taken to a skilled service man for attention.

When should water be added to the Battery? *Answer No. 135*

Add nothing but pure water to the cells and do it often enough to keep the plates covered at all times. The solution (electrolyte) should be maintained at a level with the bottom of the filling tube. Distilled water, melted artificial (but not natural) ice and rain water, if obtained in the open country from a clean slate or shingle-covered roof, are generally satisfactory. All water for battery use should be kept in clean, covered vessels of glass, china, earthenware, rubber or lead. In cold weather add water only immediately before running the engine so that the charging will mix the water and electrolyte, and avoid freezing. If, for any reason, it is necessary to add acid, the battery should be taken to an authorized Service Station.

What care should be given the Filling Plugs and Connections? *Answer No. 136*

Keep the filling plugs and connections tight and the top of the battery clean. Wiping the battery with a rag moistened with ammonia will counteract the effect of any of the solution which may be on the outside of the battery. A coating of heavy oil or vaseline will protect the connectors from corrosion. Keep the battery firmly secured in position. If clamps are loose, the battery will shift about in the compartment and result in loose connections, broken cells and other trouble. If repairs are necessary or if the car is to be laid up for the winter, take the battery to a skilled service man for proper attention and storage. Do not entrust the battery to inexperienced or unskilled hands.

Summary of Engine Troubles and Their Causes

ENGINE FAILS TO START

1. Gas mixture too lean.
2. Water in gasoline.
3. Vibrators adjusted too close.
4. Water or congealed oil in commutator.
5. Magneto contact point (in trans. cover) obstructed with foreign matter.
6. Gasoline supply shut off.
7. Carburetor frozen (in zero weather).
8. Water frozen in gasoline tank sediment bulb.
9. Coil switch off.

ENGINE LACKS POWER—RUNS IRREGULARLY

At Low Speeds.

1. Poor compression—account leaky valves.
2. Gas mixture too rich or too lean.
3. Spark plugs dirty.
4. Coil vibrator improperly adjusted.
5. Air leak in intake manifold.
6. Weak exhaust valve spring.
7. Too great clearance between valve stem and push rod.
8. Too close gap between spark plug points.

At High Speeds.

1. Commutator contact imperfect.
2. Weak valve spring.
3. Too much gap in spark plug.
4. Imperfect gas mixture.
5. Vibrator points dirty or burned.

ENGINE STOPS SUDDENLY

1. Gasoline tank empty.
2. Water in gasoline.
3. Flooded carburetor.
4. Dirt in carburetor or feed pipe.
5. Magneto wire loose at either terminal.
6. Magneto contact point obstructed.
7. Overheated—account lack of oil or water.
8. Gas mixture too lean.

ENGINE OVERHEATS

1. Lack of water.
2. Lack of oil.
3. Fan belt torn, loose or slipping.
4. Carbon deposit in combustion chamber.
5. Spark retarded too far.
6. Gas mixture too rich.
7. Water circulation retarded by sediment in radiator.
8. Dirty spark plugs.

ENGINE KNOCKS

1. Carbon deposit on piston heads.
2. Loose connecting rod bearing.
3. Loose crank shaft bearing.
4. Spark advanced too far.
5. Engine overheated.